# Linux Operating System

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# Today's Agenda

01 Why use Linux?

02 How to install Linux?

03 How to use Linux?

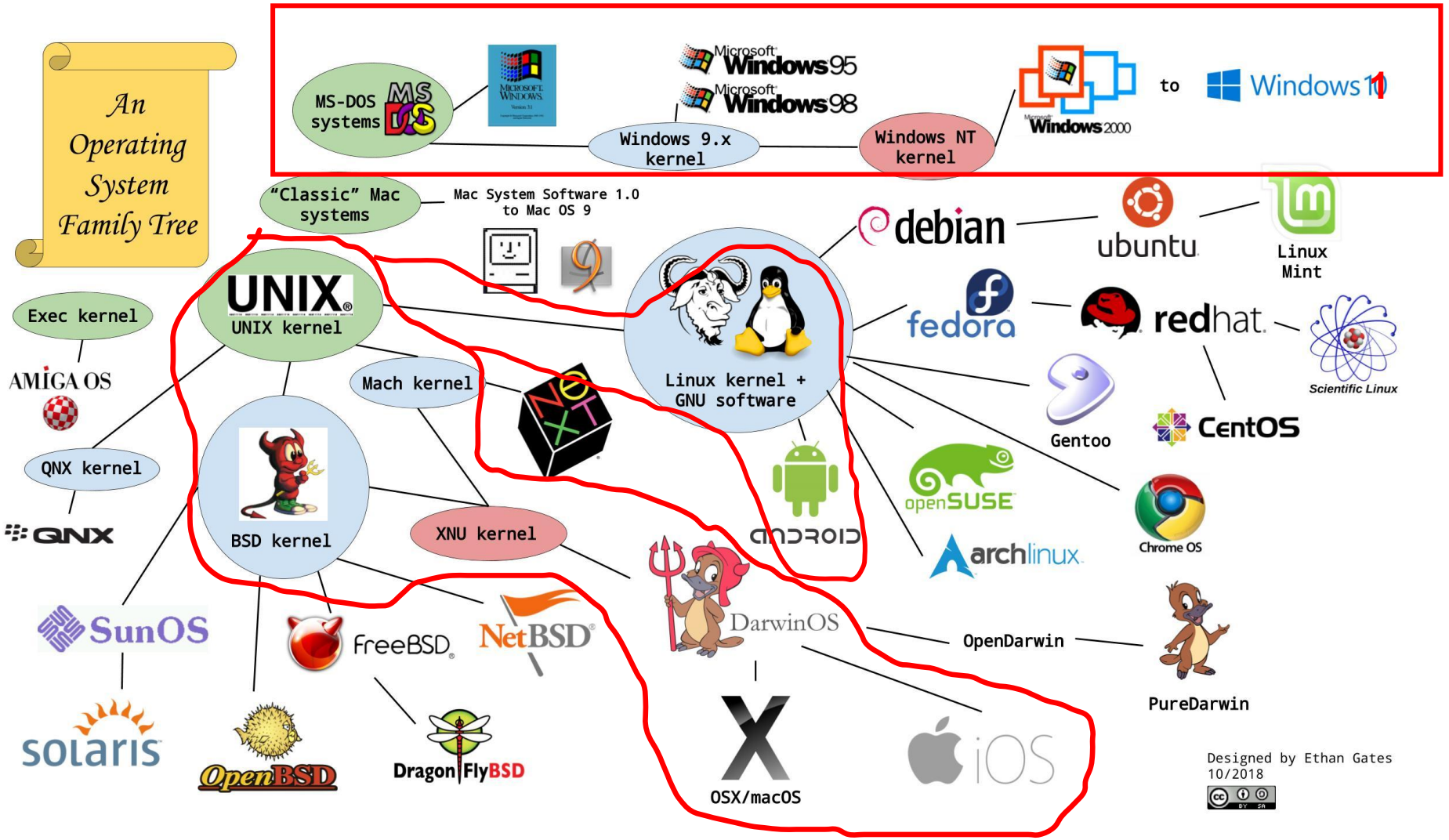
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# Operating Systems



# Operating Systems

Commercially successful operating systems




- **Advantages**
  - Easy to use
  - Many applications
  - Stable and reliable
- **Disadvantages**
  - Limited control over the system
  - Dependence on commercial ecosystems

# Operating Systems

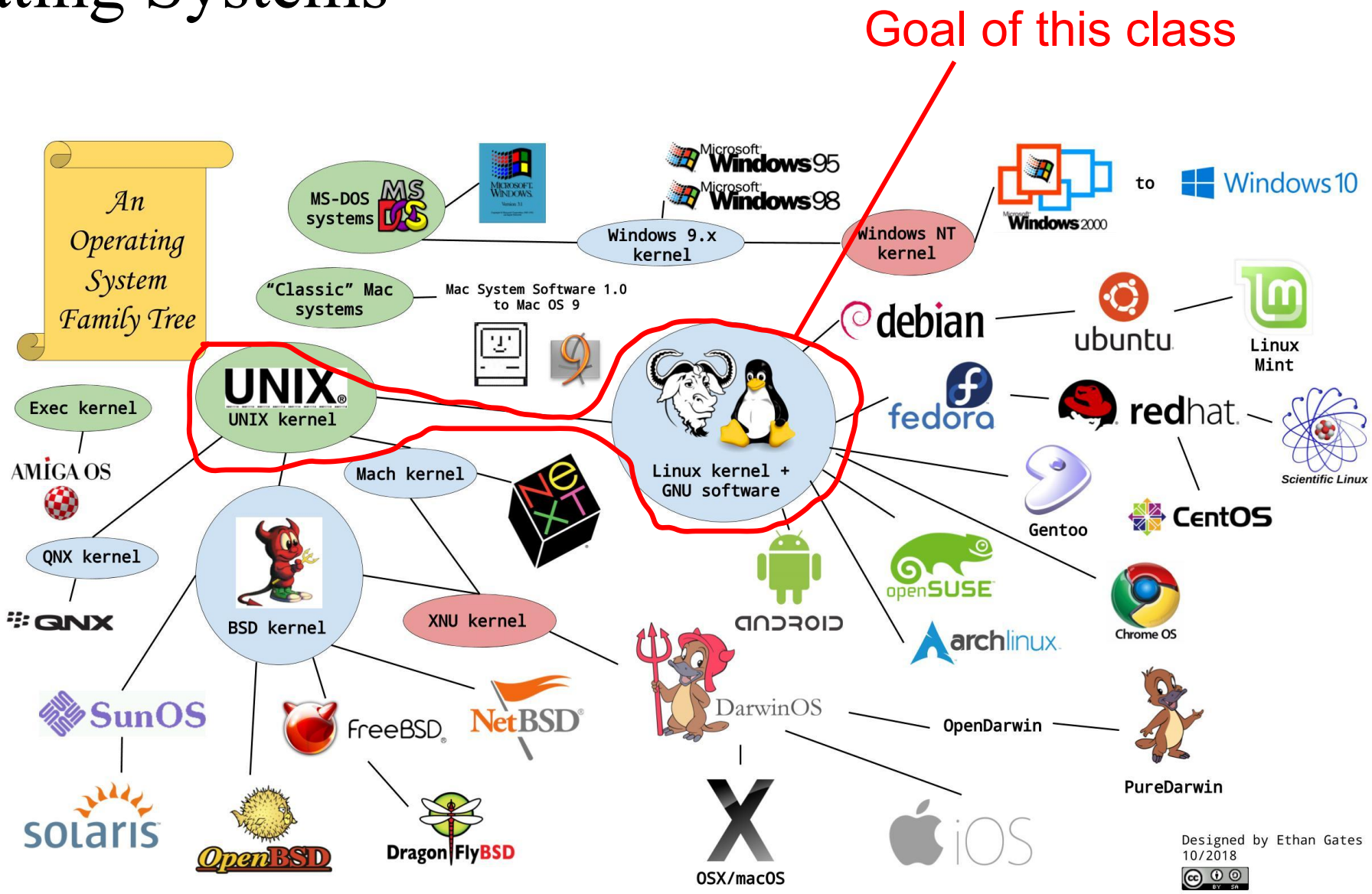


**When you want to develop software or do research...**

Linux is often a great choice because it is:

- **Free and open-source**
- **Flexible and customizable**
- **Widely used on servers and in research**

# Operating Systems



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# Installation (Windows)



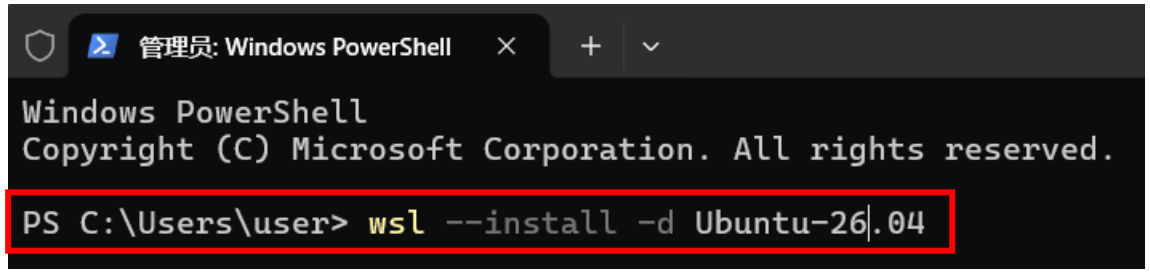

1) Right-click

2) Click

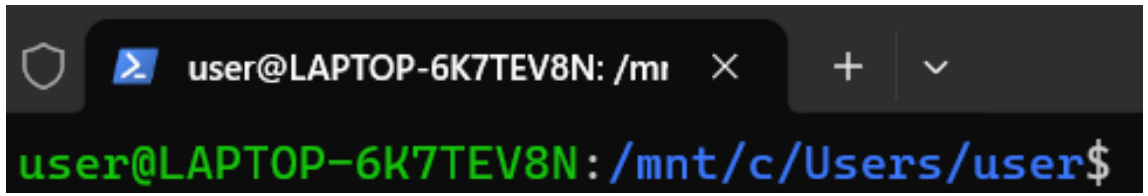


# Installation (Windows)

3) Enter this command and press “Enter”, then follow its instructions.

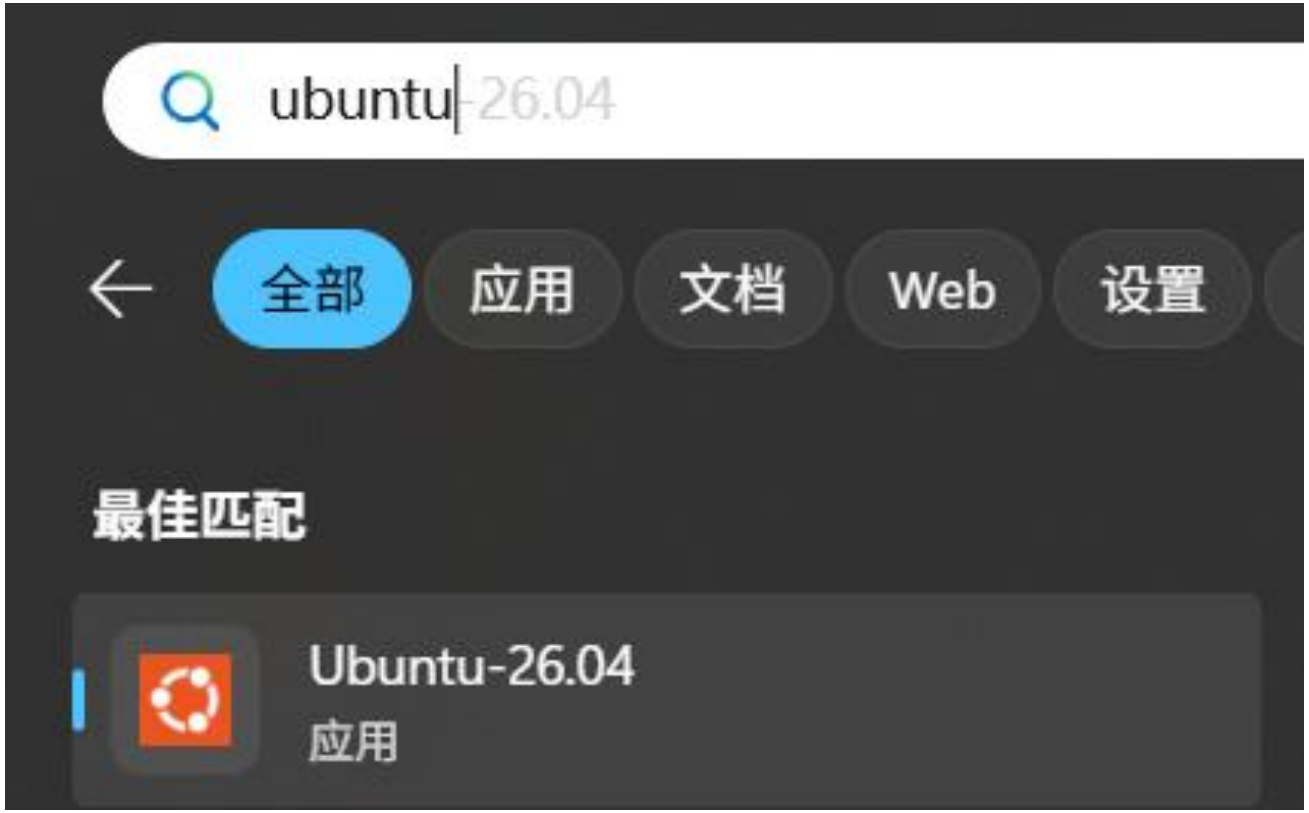



4) If you see sth like this, you succeed.



# Installation (Windows)

After that, you can find it in your start menu.



# Installation (MacOS)

MacOS is a Unix-based operating system, so you don't need to install Linux for this class. You can simply use the Terminal on your Mac.

# Today's Agenda

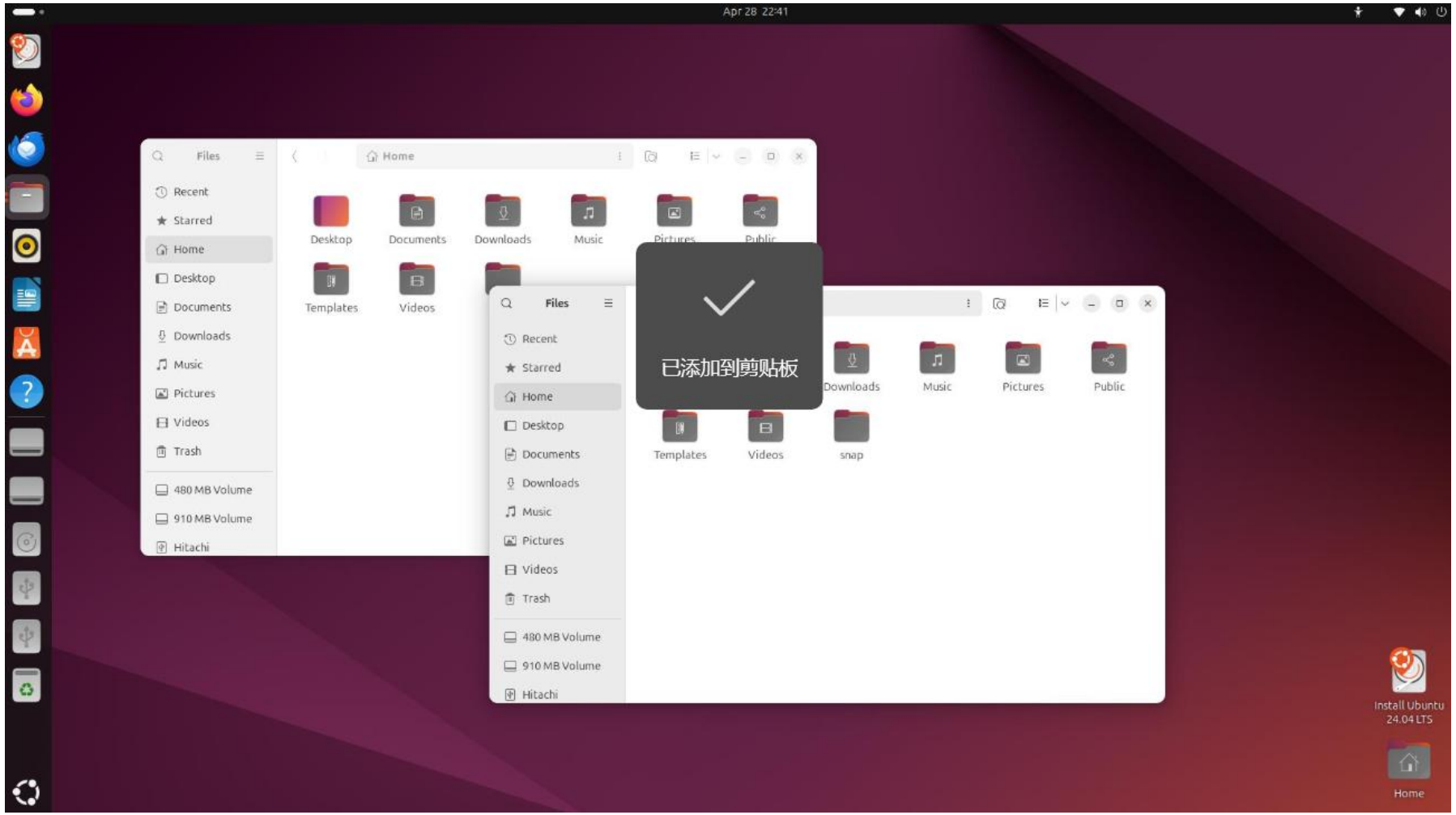
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# GUI

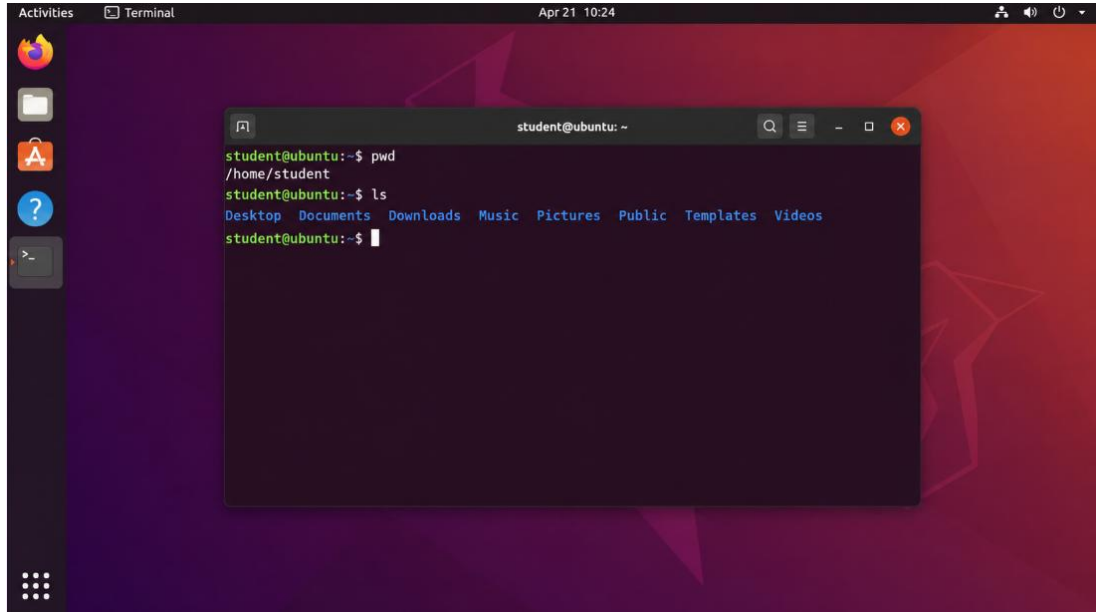
Most Linux distributions also provide a graphical user interface (GUI).



# CLI

A command-line interface (CLI) allows users to interact with a computer by typing text commands instead of using a graphical interface.

## CLI in Ubuntu system




## CLI in WSL



# GUI vs CLI

## GUI — Graphical User Interface

**Advantages**
Easy to learn and visually intuitive.

**Best for**
Visual and interactive tasks.

**Limitations**
Repetitive tasks may require many clicks.

## CLI — Command-Line Interface

**Advantages**
Fast, precise, and easy to automate.

**Best for**
Running programs and managing tools.

**Limitations**
Requires learning commands.

## Reason to learn CLI

# CLI and the File System

The CLI lets us navigate and manage files and directories using text commands.

## Example file system

```
~  (/home/user)
├── source
│   └── v1
│       ├── Documents
│       └── Programs
└── configs
```

## Common commands

| | |
|---|---|
| `pwd` | Where am I? |
| `ls` | What is in this directory? |
| `cd Documents` | Go to another directory |
| `mkdir projects` | Create a directory |
| `cp / mv / rm` | Copy, move, or remove files |

**SAFETY!!!**

A command operates relative to your current directory unless you specify another path.

# Command-Line Safety

These commands can overwrite or permanently delete files.

| | |
|---|---|
| `cp old.txt report.txt` | Overwrite an existing report.txt |
| `mv draft.txt report.txt` | Overwrite an existing report.txt |
| `rm -r project/` | Deletes the folder and everything inside it |

**Warning: In WSL, files under /mnt/ are real Windows files.**

# Files and Directories in Linux

In Linux, a directory is a special type of file.

## Useful commands

| | |
|---|---|
| `ls -lh` | List files and their sizes |
| `du -sh folder/` | Show the total disk usage of a folder |
| `stat file.txt` | Show detailed file information |

# Text File Editting

nano is a simple text editor you can use in the terminal.

## Open (or create) a text file:

```
nano notes.txt
```

---

## Edit the text, then use these shortcuts:

| | |
|---|---|
| **`Ctrl + O`** | Save the file |
| **`Enter`** | Confirm the filename |
| **`Ctrl + X`** | Exit nano |

# Installing Software in Ubuntu

APT is Ubuntu's built-in package manager for installing, updating, and removing software.

**1. Update the list of available software**

```
sudo apt update
```

**2. Install a software package (e.g., Git)**

```
sudo apt install git
```

sudo = run the command with administrator privileges

Note: Ubuntu may ask for your password.

# Where are the Softwares Installed?

## Example: Git

| | |
|---|---|
| **`/usr/bin/git`** | Executable program |
| **`/usr/share/doc/git`** | Documentation |

---

## Find the installed files:

| | |
|---|---|
| `which git` | Find the executable |
| `dpkg -L git` | List files in the package |

# How to Use Installed Software

Each program defines its own commands and options.

## Common pattern

```
program  [command]  [options]  [arguments]
```

---

## Example: Git

| | |
|---|---|
| `git status` | Check repository status |
| `git --help` | Show usage information |

# Running and Stopping Programs

A command can start a program that keeps running.

| | |
|---|---|
| `sleep 30` | Waits for 30 seconds. |
| `Ctrl + C` | Interrupts the running program. |

Try it: run sleep 30, then press Ctrl + C.

# Checking and Stopping Processes

A process is a running instance of a program.

| | |
|---|---|
| `ps aux` | Lists processes and their IDs (PIDs). |
| `kill 12345` | Requests that process 12345 stop. |

PID = Process ID (a number identifying a process).

Only stop processes you recognize.

# Checking Computer Resource Usage

Check which resources your computer is using.

| | |
|---|---|
| `top` | Live process list and CPU / memory usage. |
| `free -h` | Shows memory usage. |
| `df -h` | Shows disk space usage. |

Tip: Press q to exit top.